

Bacterial Sodium-Coupled Proline Uptake and Fused-PutA Catabolism

A commissioned review-style synthesis of a reusable bacterial proline-to-glutamate module

1. Executive Summary

Many bacteria treat the imino acid L-proline as an interchangeable source of carbon, nitrogen, and reducing equivalents. They do so with a compact, genetically coupled two-component module. First, a **PutP**-family sodium/proline symporter uses the inwardly directed electrochemical Na^+ gradient to concentrate cytoplasmic L-proline against its own gradient. Second, a single **PutA** polypeptide oxidizes that proline all the way to L-glutamate in two chemically distinct steps, using an FAD-dependent **proline dehydrogenase (PRODH)** domain and an NAD^+ -dependent **glutamate- γ -semialdehyde dehydrogenase (GSALDH, also called P5CDH)** domain. The PRODH domain passes proline-derived electrons directly to the membrane quinone pool while producing Δ^1 -pyrroline-5-carboxylate (P5C); P5C then undergoes non-enzymatic ring opening to L-glutamate- γ -semialdehyde (GSA), which the GSALDH domain oxidizes to glutamate. The reactive P5C/GSA intermediate is not simply released to bulk solvent — it is **substrate-channeled** ~ 42 Å through a buried intramolecular tunnel between the two active sites.

The defining biological insight is that this is a *modular, reusable* device assembled from two ancient and independently conserved catalytic cores. The PRODH core is a distorted $(\beta\alpha)_8$ TIM-barrel; the GSALDH core belongs to the aldehyde dehydrogenase (ALDH) superfamily. In eukaryotes and some bacteria these two activities are carried by *separate monofunctional enzymes*; in many Gram-negative bacteria they are *fused* into bifunctional PutA. In enterobacteria such as *Escherichia coli*, an additional N-terminal ribbon-helix-helix DNA-binding domain converts PutA into a *trifunctional* protein that both catalyzes proline oxidation and autoregulates the *put* regulon encoding PutA and PutP. This DNA-binding/regulatory switch is a lineage-specific elaboration layered onto the conserved bifunctional core, and it is controlled by the redox state of the PRODH flavin.

Across three investigative iterations, eight findings were confirmed with structural, kinetic, and reconstitution evidence drawn from *E. coli*, *Salmonella typhimurium*, *Bradyrhizobium japonicum*, *Thermus thermophilus*, and comparative eukaryotic systems. The system's boundaries are sharp: it should be clearly distinguished from the mammalian PRODH/POX redox-signaling axis (a p53-regulated, mitochondrial, tumor-suppressor pathway that performs analogous chemistry in a completely different regulatory and cell-biological context) and from proline *biosynthesis* and the proline–P5C cycle. The best current model is a linear, obligately ordered pathway — uptake → PRODH oxidation → non-enzymatic ring opening → GSALDH oxidation — in which channeling and (in some lineages) transcriptional autoregulation are the key layers of sophistication built on top of two very old enzymes.

2. Definition and Biological Boundaries

2.1 What is included

The system, as scoped here, comprises three obligatory chemical/transport events and the molecular machines that carry them out:

1. **Sodium-coupled proline uptake** by a PutP-family Na⁺/proline symporter (solute:sodium symporter, SSS, family), which delivers cytoplasmic L-proline.
2. **Quinone-linked proline oxidation** by the FAD-dependent PRODH domain of PutA, generating P5C and reduced quinol.
3. **Glutamate-semialdehyde oxidation** by the NAD⁺-dependent GSALDH/P5CDH domain of PutA, generating L-glutamate and NADH, after non-enzymatic P5C→GSA ring opening.

In enterobacterial variants, a fourth, *accessory* function — transcriptional autoregulation of the *put* regulon by the DNA-binding domain of trifunctional PutA — is genetically part of the same module, because the regulon it controls encodes both PutP and PutA.

2.2 What should be treated separately (neighboring/confusable processes)

- **Eukaryotic PRODH/POX signaling.** In mammals, proline oxidation is performed by monofunctional PRODH/POX, a p53-activated mitochondrial enzyme that donates electrons to the electron transport chain to generate reactive oxygen species (ROS) or ATP and acts as a **tumor suppressor** initiating apoptosis and autophagy ([PMID: 22886911](#); [PMID: 28990419](#)). This is chemically analogous but regulatorily and cell-biologically distinct: it is not fused to

GSALDH, is regulated by p53, c-MYC (via miR-23b), and PPAR γ , and participates in a proline–P5C cycle interlocked with the pentose phosphate pathway. It should not* be merged with the bacterial catabolic module.

- **Proline biosynthesis and the proline–P5C cycle.** Synthesis of proline from glutamate or ornithine (via P5C synthase/P5C reductase, or ornithine δ -aminotransferase) shares the P5C intermediate but runs in the opposite direction and is enzymatically separate. In plants and other eukaryotes, the balance between catabolism and biosynthesis, and cycling of P5C, is a distinct regulatory problem.
- **The general Na⁺/substrate symport family.** PutP is one member of a broad class of bacterial Na⁺-cotransporters (proline, glutamate, serine/threonine, citrate, branched-chain amino acids, melibiose). PutP's specificity for proline, and its role in feeding PutA, is what makes it part of *this* module; the broader family should not be conflated with it.
- **Osmoprotective proline transport.** Proline can also be accumulated as a compatible solute for osmotic stress (via other transporters such as ProP/ProU in *E. coli*). That osmoprotection role is physiologically distinct from PutP's catabolic-feeder role, even though both move proline.

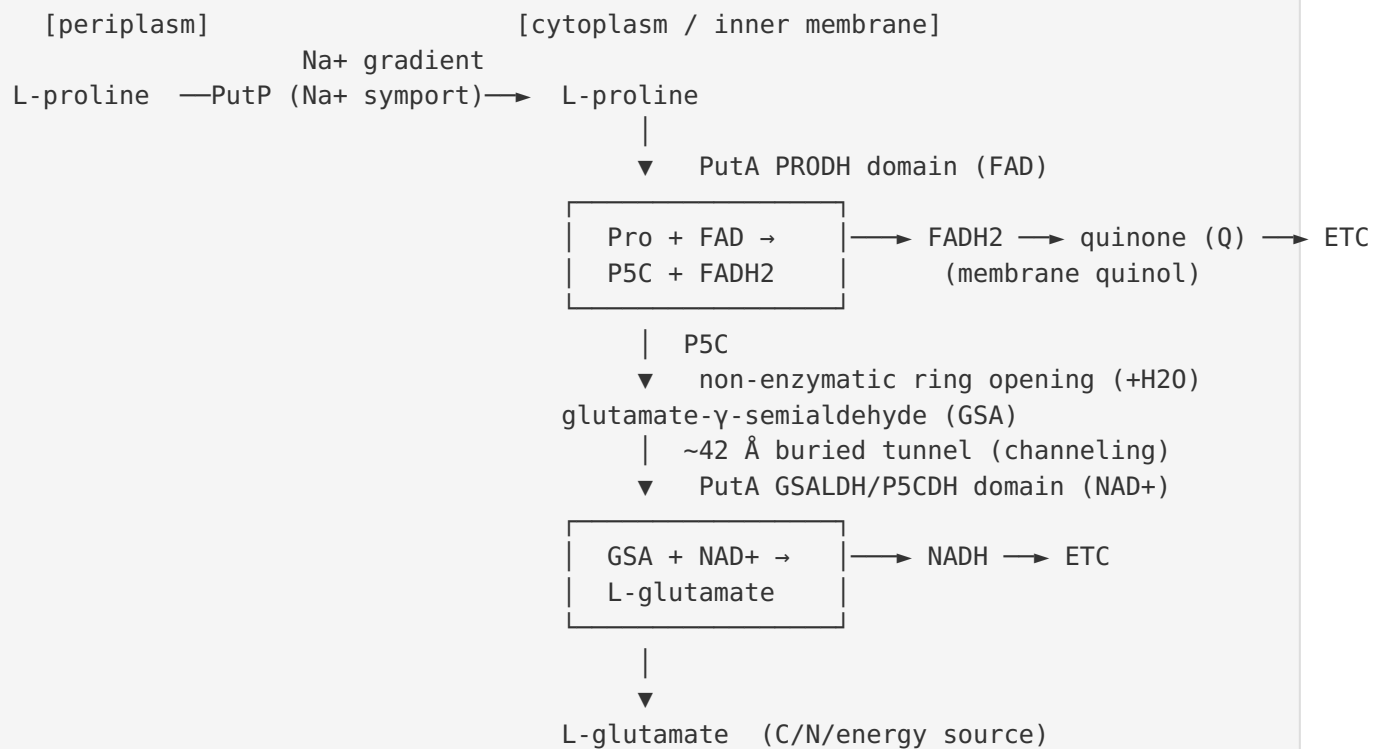
2.3 Competing definitions

The main definitional variation in the literature concerns **how many functions PutA has**. "PutA" refers to bifunctional (PRODH + GSALDH) enzymes in many Gram-negative bacteria (e.g., *B. japonicum*, *P. putida* PSEPK) and to trifunctional (DNA-binding + PRODH + GSALDH) enzymes in enterobacteria (*E. coli*, *S. typhimurium*). Both are legitimately "PutA," and a bifunctional PutA can be experimentally converted into a trifunctional one by fusing on a DNA-binding domain (PMID: 27742866). The scope brief anchors on the *bifunctional* PSEPK-type core as the canonical "reusable module," treating the DNA-binding domain as an add-on.

3. Mechanistic Overview

3.1 The best current model

The pathway is linear and obligately ordered:



3.2 Obligatory, conditional, and accessory steps

Step	Status	Rationale
PutP Na ⁺ /proline symport	Obligatory (for using external proline)	Supplies cytoplasmic substrate; proline is polar and needs a transporter
PRODH (FAD) oxidation → P5C	Obligatory	First committed catabolic step; reduces FAD
Electron transfer FADH ₂ → quinone	Obligatory (for catabolic turnover & energy capture)	Regenerates oxidized FAD; couples to respiration
Non-enzymatic P5C ⇌ GSA ring opening	Obligatory	Spontaneous; provides the GSALDH substrate
GSALDH (NAD ⁺) oxidation → glutamate	Obligatory	Completes conversion to glutamate; yields NADH
Substrate channeling of P5C/GSA	Conditional/accessory	Improves efficiency and protects the reactive intermediate, but the chemistry can proceed with free intermediate
DNA-binding autoregulation	Accessory / lineage-specific	Present only in trifunctional (enterobacterial) PutAs
Redox-driven membrane association switch	Accessory / lineage-specific	Couples regulation to catalytic state in trifunctional PutAs

3.3 Molecular assemblies carrying out each step

- **Uptake:** PutP, a LeuT-fold, SSS-family secondary transporter, working by alternating access.
- **Oxidation step 1:** the PutA PRODH domain, a distorted (β α)₈ TIM barrel binding non-covalent FAD, physically associated with the inner membrane so electrons can reach ubiquinone.
- **Intermediate handling:** a buried intramolecular tunnel (~42 Å) linking the two catalytic centers within one PutA polypeptide (or, in *T. thermophilus*, a transient PRODH–P5CDH protein–protein complex).

- **Oxidation step 2:** the PutA GSALDH/P5CDH domain, an ALDH-superfamily fold binding NAD⁺.

4. Major Molecular Players and Active Assemblies

4.1 PutP — the sodium/proline symporter

Finding F005 establishes PutP as a **LeuT-fold Na⁺/proline symporter of the SSS family**. The core transmembrane helix TM6' is central to substrate binding and translocation ([PMID: 27793991](#)), and external loop 4 (eL4) together with the outer end of TM10' forms the outer gate that occludes the substrate site during the alternating-access cycle ([PMID: 26728461](#)). Substrate and Na⁺ binding are **electrogenic and ordered**: charge-translocation measurements on a solid-supported membrane show that both Na⁺ and the *uncharged* cosubstrate proline induce charge displacement, that the two sites interact cooperatively, and that proline binding proceeds in a two-step (low-affinity electroneutral binding followed by a near-irreversible electrogenic conformational transition) process ([PMID: 15476811](#)). SSS members including PutP exhibit a **two-substrate-site (stoichiometry 2) binding architecture** consistent with a second substrate-binding site required for translocation ([PMID: 25398883](#)). Cys-scanning mutagenesis of individual transmembrane domains (e.g., TM II, TM IX) has mapped residues forming ligand-sensitive hydrophilic cavities that reconstitute parts of the Na⁺/proline translocation pathway ([PMID: 12923181](#); [PMID: 18692508](#)).

Functionally, PutP is the module's **feeder**: it converts the cell's electrochemical Na⁺ gradient into a concentrative proline supply for PutA. Bacterial Na⁺/substrate cotransport, of which PutP proline transport is a classic example, couples downhill Na⁺ entry to uphill substrate accumulation ([PMID: 11248194](#)).

4.2 PutA — the bifunctional (or trifunctional) flavoenzyme

Finding F001 — PutA is a **bifunctional flavoenzyme** that oxidizes L-proline to L-glutamate using **spatially separated PRODH and GSALDH active sites** ([PMID: 29295473](#)). Structural work places the two active sites **42 Å apart, connected by a buried tunnel** hypothesized to channel intermediates ([PMID: 40738191](#)). This physical separation is the structural basis for the module's most elegant feature.

Finding F002 — the P5C/GSA intermediate is **substrate-channeled**. Kinetic (approach-to-steady-state NADH burst), mutational (tunnel-constricting D779Y/W mutants in *B. japonicum* PutA), and structural evidence converge on an intramolecular channel that promotes channeling of the reactive intermediate (PMID: 25046425). Importantly, channeling is not unique to the fused enzyme: in *Thermus thermophilus*, the *separate* monofunctional PRODH and P5CDH physically associate (with a validated protein–protein interaction predicted by domain-fusion analysis) to channel the same intermediate (PMID: 25492892). This tells us channeling is a *conserved solution to a chemical problem* (protecting a labile, potentially toxic intermediate), realized either by domain fusion or by complex formation.

Finding F004 — the PRODH domain **passes electrons directly to the membrane quinone pool**. In *S. typhimurium*/*E. coli*, PutA proline dehydrogenase activity is efficiently reconstituted with liposomes containing ubiquinone and cytochrome *bo*, confirming that PRODH transfers electrons from reduced FAD directly to the quinone pool (PMID: 10561467). FAD reduction — not electron transfer per se — is the trigger for the conformational change underlying membrane association. This links the catabolic chemistry directly to respiration and to the regulatory switch in trifunctional PutAs.

Finding F003 — trifunctional PutAs (e.g., *E. coli*) **switch between a DNA-binding repressor and a membrane-bound enzyme under flavin redox control**. EcPutA carries an N-terminal DNA-binding domain in addition to PRODH and GSALDH; proline oxidation reduces the FAD, triggering a conformational change that releases PutA from *put* regulon DNA and promotes membrane association via a conserved C-terminal motif (PMID: 29090935). A bifunctional PutA can be engineered into a trifunctional one by fusing on a DNA-binding domain, demonstrating the modular, "add-on" nature of the regulatory function (PMID: 27742866).

4.3 The genetic module: the *put* regulon

Finding F006 — in enterobacteria, trifunctional PutA **autoregulates the *put* regulon encoding both PutA and PutP**. As a regulatory protein, EcPutA represses transcription of the *put* regulon, which contains the genes for PutA and the proline transporter PutP; proline binding to the PRODH active site and subsequent flavin reduction induce high-affinity membrane association and relieve repression, switching PutA from a regulatory to an enzymatic role (PMID: 23713611). Conserved β 3- α 3 loop residues (Asp370/Glu372 in EcPutA) mediate the proline-triggered allosteric membrane-binding switch. This is the feature that makes the two components a genuinely *coupled module*: the same protein that consumes proline also controls the abundance of the transporter that supplies it, with the switch tied to substrate availability via flavin redox state.

5. Evolutionary and Cell-Biological Variation

5.1 The conserved catalytic cores

Finding F008 — the PRODH catalytic core is a **conserved distorted ($\beta\alpha$)₈ barrel** shared by monofunctional PRODH and the PutA PRODH domain. The 2.0-Å structure of monofunctional *T. thermophilus* PRODH reveals this barrel with nine conserved sequence motifs for cofactor/substrate recognition (FAD midpoint potential -75 mV; Km(proline) 27 mM; kcat 13 s^{-1}) ([PMID: 17344208](#)). Critically, the same paper establishes that PRODH and P5CDH are **distinct monofunctional enzymes in all eukaryotes and some bacteria, but fused into bifunctional PutA in other bacteria** — the distributional fact that underpins the domain-fusion evolutionary model.

5.2 Fusion state across lineages

Lineage / organism	PRODH-GSALDH state	Regulatory add-on	Notes
Many Gram-negative bacteria (<i>B. japonicum</i> , <i>P. putida</i> PSEPK)	Fused, bifunctional PutA	None	The canonical "reusable module"
Enterobacteria (<i>E. coli</i> , <i>S. typhimurium</i>)	Fused, trifunctional PutA	N-terminal DNA-binding domain	Autoregulates <i>put</i> regulon; redox switch
<i>Thermus thermophilus</i>	Separate monofunctional PRODH + P5CDH	None	Enzymes physically associate; substrate channeling via complex (Kd $\sim 3\text{ }\mu\text{M}$)
Eukaryotes (plants, fungi, protists, mammals)	Separate monofunctional ProDH/PRODH + P5CDH	Diverse, organism-specific	Mitochondrial; not fused; see §5.3

The **fusion is best interpreted as a bacterial elaboration** of two pre-existing, independently folded enzymes rather than the ancestral state. The domain-fusion analysis validated by the *T. thermophilus* PRODH–P5CDH interaction ([PMID: 25492892](#)) is a textbook case: proteins fused in one lineage tend to physically interact in lineages where they remain separate, because they participate in the same reaction sequence. Because both the PRODH ($\beta\alpha$)₈ barrel and the ALDH-superfamily GSALDH exist as standalone enzymes, the **best representatives for understanding the ancestral role** are the monofunctional forms (e.g., *T. thermophilus* PRODH), while the fused PutA represents the derived, integrated device.

5.3 Eukaryotic variation and the boundary of the system

Finding F007 delineates the eukaryotic boundary. In mammals, proline oxidation is a **p53-regulated redox-signaling axis** distinct from bacterial fused PutA: PRODH/POX donates electrons to the ETC to generate ROS or ATP, acting as a tumor suppressor that initiates apoptosis/autophagy, and is regulated by c-MYC (via miR-23b), *PPAR γ* , and *metabolic context*, participating in a proline–P5C cycle interlocked with the pentose phosphate pathway ([PMID: 22886911](#); [PMID: 28990419](#); [PMID: 26598224](#); [PMID: 22615405](#)). In plants, *ProDH* and *P5CDH* are separate mitochondrial enzymes; *ProDH* contributes to pathogen defense and the hypersensitive response, and can drive a ROS-generating Pro–P5C cycle when *P5CDH* is limiting ([PMID: 21311034](#); [PMID: 26284090](#)). Notably, plant mitochondria assemble a physical P5C-metabolism complex (*ProDH* + *P5CDH* + ornithine δ -aminotransferase) that may channel P5C and feed electrons to the respiratory chain — a eukaryotic echo of the bacterial channeling solution, but without fusion ([PMID: 37843921](#)). In fungi (*Cryptosporidia parasitica*) and protists (*Trypanosoma cruzi**), the two-step proline→glutamate oxidation is essential for virulence/differentiation and feeds reducing equivalents to the mitochondrial ETC ([PMID: 24039956](#); [PMID: 34661234](#)).

The common thread across all lineages is the *chemistry* (two oxidations, one non-enzymatic ring opening, electron delivery to a quinone/ETC). What varies is **architecture** (fused vs. separate vs. complexed), **compartment** (bacterial cytoplasm/inner membrane vs. eukaryotic mitochondrion), and **regulation** (put-regulon autoregulation vs. p53/MYC signaling vs. senescence/defense programs).

6. Constraints, Dependencies, and Failure Modes

6.1 Ordering constraints

- **Uptake precedes oxidation.** External proline must be imported before PutA can act (unless the cell uses endogenous proline). PutP is therefore upstream and rate-influencing.
- **PRODH precedes GSALDH.** GSA (the GSALDH substrate) does not exist until PRODH has made P5C and it has ring-opened. This ordering is enforced by chemistry, not just by enzyme organization.
- **Ring opening is spontaneous and interposed.** The $P5C \rightleftharpoons GSA$ equilibrium is non-enzymatic; the enzyme does not catalyze it but exploits it. This is why the intermediate is described as "P5C/GSA."
- **Flavin reduction precedes the regulatory switch.** In trifunctional PutAs, the conformational change that drives membrane association and derepression requires FAD reduction, which requires proline. Thus the regulatory switch cannot fire without substrate — a built-in coherence.

6.2 Compartment/context specificity

- PutA's catalytic turnover requires **access to the membrane quinone pool** to reoxidize FAD; the reconstitution with ubiquinone/cytochrome *bo* liposomes demonstrates this dependency ([PMID: 10561467](#)). A soluble PutA disconnected from the membrane cannot sustain catabolic flux.
- PutP function depends on an **inward Na^+ electrochemical gradient**; collapse of the gradient abolishes concentrative uptake.

6.3 Failure modes and evidence ruling out alternative paths

- **Loss of channeling.** Tunnel-constricting mutations (D779Y/W in BjPutA) perturb kinetics, showing that the physical tunnel matters for efficient handoff ([PMID: 25046425](#)). This rules out a model in which the intermediate is freely and efficiently exchanged with bulk solvent as the primary route.
- **Substrate inhibition.** Excess proline can inhibit PutA; a structural basis for substrate inhibition of PutA by proline has been described ([PMID: 29295473](#)), a physiological constraint on turnover at high substrate.

- **Flavin modification / inhibition.** The PRODH flavin can be covalently modified (e.g., by thiazolidine-2-carboxylate), inactivating the enzyme — a chemical vulnerability of the FAD-dependent step (PMID: 32159324).
- **Reactive-intermediate toxicity.** P5C/GSA is reactive; unchanneled accumulation is deleterious. In plants, blocking P5CDH while ProDH is active drives a ROS-generating Pro-P5C cycle (PMID: 21311034), illustrating what happens when the second oxidation cannot keep pace — a failure mode the bacterial fused/channeled architecture is well suited to avoid.

7. Controversies and Open Questions

Strongly supported claims. The bifunctional architecture with two spatially separated active sites (F001), direct electron transfer from PRODH FAD to the quinone pool (F004), the redox-driven regulatory switch in trifunctional PutAs (F003, F006), the conserved ($\beta\alpha$)₈ PRODH barrel (F008), and PutP's SSS/LeuT-fold identity and two-site electrogenic binding (F005) are all backed by direct structural, kinetic, mutational, and/or reconstitution data.

Areas of indirect evidence or debate.

1. **How complete is channeling?** Channeling is supported by kinetics, structure, and mutagenesis (PMID: 25046425; PMID: 40738191), but the *fraction* of intermediate channeled vs. released, and how this varies among PutAs and physiological conditions, is not fully quantified. The tunnel is "hypothesized to channel" the intermediate — a model strongly consistent with, but not identical to, a fully proven quantitative flux measurement.
2. **Cross-organism extrapolation.** Structural and mechanistic detail is concentrated in a few model PutAs (EcPutA, BjPutA) and in *T. thermophilus* for the separate-enzyme case. The brief's nominal target, *P. putida* PSEPK PutA, is inferred to share the conserved bifunctional PRODH–GSALDH core but has not itself been the subject of the deepest mechanistic studies cited here. Claims about PSEPK specifically rest on conservation arguments.
3. **PutP–PutA physical/functional coupling.** The two are co-regulated genetically (F006), and PutA associates with the membrane where PutP resides, but whether there is any direct physical or kinetic coupling (e.g., proline handoff microdomains) is unresolved.
4. **PutP transport stoichiometry and second-site details.** The two-substrate-site model (PMID: 25398883) and electrogenic ordered binding (PMID: 15476811) are supported, but a

high-resolution PutP structure resolving both Na⁺ sites and the proline site simultaneously remains a gap (much inference is by analogy to LeuT).

5. **Boundary conflation risk.** The literature frequently mixes bacterial PutA data with eukaryotic PRODH/POX signaling. These should not be pooled; the regulatory logic (put-regulon autoregulation vs. p53/MYC) and cell biology (bacterial membrane vs. mitochondrion) are fundamentally different (F007).

Most important open questions. - Quantitative channeling efficiency and its physiological tuning across PutA homologs. - A direct high-resolution structure of PSEPK PutA (and of PutP with all ligands) to test conservation assumptions. - Whether/how PutP abundance and PutA membrane association are spatially or kinetically coordinated in vivo. - The evolutionary trajectory: order of domain acquisition (PRODH+GSALDH fusion first, DNA-binding domain later) and the selective advantage of fusion vs. transient complex formation.

8. Mechanistic Model and Interpretation

Synthesizing the eight findings: **the bacterial proline-utilization module is a domain-fusion device that solves three coupled problems — substrate acquisition, safe two-step oxidation of a reactive intermediate, and demand-matched gene regulation — by combining two ancient enzymes with a dedicated transporter under a single genetic switch.**

- The **transporter (PutP)** converts an ion gradient into concentrative proline supply (F005).
- The **fusion (PutA)** places two independently ancient catalytic cores — a (β α)₈-barrel PRODH (F008) and an ALDH-superfamily GSALDH — in one polypeptide, ~42 Å apart, joined by a tunnel that channels the labile P5C/GSA intermediate (F001, F002).
- The **first oxidation** is wired to respiration by direct FADH₂→quinone electron transfer (F004), which simultaneously provides the redox signal.
- That redox signal drives, in enterobacteria, a **regulatory switch** (F003) that couples catalysis to autoregulation of the very regulon encoding both PutA and PutP (F006).
- The whole device is a bacterial elaboration; the same chemistry runs on *separate* enzymes (sometimes physically complexed, as in *T. thermophilus* and plant mitochondria) in other lineages, and on a p53-governed mitochondrial enzyme in animals (F007, F008).

The elegance is that fusion converts what would be three loosely coupled activities (import, oxidize, oxidize) into a coordinated, self-regulating unit whose regulatory state reports directly on substrate flux through the flavin.

9. Limitations and Knowledge Gaps

- This review is a **literature synthesis**, not an experimental study; no new data were generated. Findings rest on the primary literature summarized above.
 - Mechanistic depth is **model-organism-biased** (EcPutA, BjPutA, StPutA, *T. thermophilus*). Statements about *P. putida* PSEPK PutA are conservation-based inferences.
 - **Quantitative parameters** (channeling fraction, in-vivo fluxes, PutP transport rates under physiological conditions) are incompletely characterized.
 - The **PutP structural model** relies substantially on homology to LeuT rather than a native high-resolution PutP structure with all ligands resolved.
 - The separation between bacterial catabolism and eukaryotic signaling is emphasized here, but some cited eukaryotic work (plants, fungi, protists, mammals) is used to *illustrate boundaries and conserved chemistry*, not to make claims about the bacterial system directly.
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10. Proposed Follow-up Experiments / Actions

1. **Solve/predict a PSEPK PutA structure** (cryo-EM or AlphaFold-Multimer + validation) and superpose it on EcPutA/BjPutA to test the assumption that the PRODH-GSALDH core and tunnel are conserved in the nominal target organism.
2. **Quantify channeling efficiency** in PSEPK vs. EcPutA/BjPutA using transient-kinetic (NADH burst) and isotope-dilution assays with tunnel mutants, to convert the "hypothesized channeling" into a measured flux fraction.
3. **Reconstitute PutP + PutA together** in proteoliposomes with a Na⁺ gradient and a quinone pool to test for functional/kinetic coupling between import and oxidation.
4. **Determine a ligand-bound PutP structure** (with both Na⁺ sites and proline) to validate the two-substrate-site, electrogenic ordered-binding model derived from LeuT analogy and SSM electrophysiology.

5. **Comparative genomics of *put* loci** across Gram-negatives to map where PutA is bi- vs. trifunctional and whether PutP is always co-regulated, clarifying the evolutionary order of domain acquisition.
 6. **In-vivo flux and stress assays** in PSEPK to test whether loss of channeling (tunnel mutants) causes P5C/GSA accumulation and ROS, as predicted by the plant P5CDH-limitation failure mode.
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11. Key References

PMID	Relevance
29295473	PutA bifunctionality; spatially separated PRODH/GSALDH sites; substrate inhibition by proline (F001)
40738191	42 Å active-site separation; buried channeling tunnel; covalent intermediates (F001, F002)
25046425	Tunnel-perturbing mutants (D779Y/W) in BjPutA support channeling (F002)
25492892	Fused-in-bacteria vs. separate-enzyme distinction; validated PRODH–P5CDH interaction (F002)
29090935	Redox-driven membrane-binding domain; repressor↔enzyme switch (F003)
27742866	Engineering trifunctional PutA by fusing a DNA-binding domain (F003)
10561467	Reconstitution with ubiquinone/cytochrome <i>bo</i> ; direct electron transfer to quinone (F004)
27793991	PutP as SSS/LeuT-fold Na ⁺ /proline symporter; TM6' role (F005)
25398883	Second substrate-binding site; stoichiometry 2 in PutP (F005)
26728461	eL4/TM10' outer gate in PutP alternating access (F005)
15476811	Electrogenic, ordered, two-step substrate binding in PutP (F005)
12923181	TM II aqueous cavity; Na ⁺ /proline binding residues in PutP
18692508	TM IX functional residues in PutP
23713611	<i>put</i> regulon encodes PutA+PutP; β3-α3 loop allosteric switch (F006)
22886911	Eukaryotic PRODH/POX feeds ETC; tumor context (F007)
28990419	p53-activated PRODH/POX redox signaling; boundary (F007)
26598224	Proline biosynthesis/PB, proline–P5C cycle, pyridine nucleotides (eukaryotic boundary)
22615405	c-MYC reprogramming of proline/glutamine metabolism (eukaryotic boundary)
17344208	<i>T. thermophilus</i> PRODH (βα) ₈ barrel; monofunctional-vs-fused distribution (F008)
28712849	Comprehensive review of PutA structure/function/mechanism
11248194	Bacterial Na ⁺ /substrate cotransport overview

PMID	Relevance
37843921	Plant mitochondrial P5C-metabolism complex (eukaryotic channeling echo)
21311034	Plant ProDH, Pro-P5C cycle, ROS failure mode
26284090	P5CDH modulation of P5C fate in plants
24039956	Fungal ProDH/P5CDH essential for virulence/sporulation
34661234	<i>T. cruzi</i> PRODH/P5CDH feed mitochondrial ETC; differentiation
32159324	Covalent flavin modification of PRODH (failure mode)

*Prepared as a commissioned review synthesis. Uncertainty is flagged throughout; claims about specific organisms (notably *P. putida* PSEPK) that rest on conservation rather than direct data are identified as such.*

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